ModelIt!

Student Activity Pack

Why Is Antibiotic Resistance Getting Worse?

Modeling Natural Selection, Mutation, and the Evolution of Resistant Bacteria

Grade 12L1 | NGSS HS-LS4-2 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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Table of Contents

ModelIt!

1 Pre-Assessment 3

2 Scoring Rubric 4

3 Component Analysis 5

4 Model Recording Page 6

5 Simulation Observations 7

6 Research & Extend 8

7 Reflection Questions 9

8 STEM Challenge 10

1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Have you or someone you know ever stopped taking antibiotics early because you felt better? What do you think happens to the bacteria that survive?**

2. **Why do you think antibiotics don't work against viruses like the common cold or flu?**

3. **What does the term 'superbug' mean to you, and why do you think they're becoming more common?**

4. **Do you think bacteria 'learn' to resist antibiotics, or is something else happening? Explain your thinking.**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Antibiotic Exposure | Bacterial Population Diversity | Mutation Rate | Resistance Gene Frequency | Treatment Effectiveness

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Complete Treatment

Antibiotic Exposure: Full dose, 10 days | Population Diversity: Moderate | Mutation Rate: Normal

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Stopped Early

Antibiotic Exposure: Full dose, 5 days then stopped | Population Diversity: Moderate | Mutation Rate: Normal

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Farm Antibiotics

Antibiotic Exposure: Low constant dose | Population Diversity: High | Mutation Rate: Normal

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Horizontal Gene Transfer Rate — The frequency at which bacteria exchange resistance genes through conjugation, transformation, and transduction — allows resistance to spread between unrelated species without reproduction

Immune System Response — The host's innate and adaptive immune defenses working alongside antibiotics — when immunity is compromised, even low levels of resistant bacteria can cause life-threatening infections

Biofilm Formation — Bacterial communities encased in a protective matrix that reduces antibiotic penetration by up to 1000x — bacteria in biofilms require much higher antibiotic concentrations to kill, promoting resistance development

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Every year, 1.27 million people die worldwide from antibiotic-resistant infections — more than HIV/AIDS or malaria. If antibiotics are one of humanity's greatest inventions, how are microscopic bacteria outsmarting our best medicines, and are we heading toward a post-antibiotic apocalypse? Use evidence from your simulation to support your answers.

1. **How does your model demonstrate that antibiotic resistance is an example of evolution by natural selection happening in real time?**

2. **Why is stopping antibiotics early one of the most dangerous things a patient can do, according to your model?**

3. **If bacteria evolve resistance to every antibiotic we create, is developing new antibiotics a long-term solution? What does your model suggest?**

4. **How does agricultural antibiotic use connect to human health, even if people never directly consume those antibiotics?**

5. **What are the ethical implications of wealthy countries having access to last-resort antibiotics while developing nations do not?**

8 STEM Challenge: Design an Antibiotic Stewardship Campaign

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YOUR ENGINEERING MISSION

Design an evidence-based public awareness and behavior change campaign that reduces antibiotic misuse in your community by targeting the specific behaviors that drive resistance evolution.

THE SCENARIO

Your county health department has documented a 40% increase in antibiotic-resistant infections over the past decade. Data shows the top three drivers are: patients demanding antibiotics for viral infections, patients not completing prescribed courses, and agricultural antibiotic overuse. Your team must design a multi-pronged campaign that changes these behaviors using evolutionary biology evidence.

Engineering Design Thinking

1. WHAT SPECIFIC BEHAVIORS DOES YOUR CAMPAIGN TARGET AND WHAT EVIDENCE-BASED STRATEGIES DO YOU USE TO CHANGE THEM?

2. HOW DOES YOUR CAMPAIGN COMMUNICATE EVOLUTIONARY BIOLOGY CONCEPTS IN WAYS NON-SCIENTISTS CAN UNDERSTAND AND ACT ON?

3. WHAT MODEL EVIDENCE SUPPORTS YOUR CLAIM THAT THESE INTERVENTIONS WILL SLOW RESISTANCE EVOLUTION?

4. HOW WOULD YOU MEASURE WHETHER YOUR CAMPAIGN ACTUALLY REDUCES ANTIBIOTIC MISUSE AND RESISTANCE RATES?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Epidemiologists and antimicrobial resistance researchers track the spread of resistant infections and develop containment strategies, earning $60,000-$120,000/year. Infectious disease physicians specialize in treating resistant infections and designing antibiotic stewardship programs, earning $200,000-$350,000/year. Pharmaceutical microbiologists developing new antibiotics earn $75,000-$140,000/year.